Lesson 1: Solving Systems by Graphing: How Many Solutions?

Two equations at once — the solution is the point where the two lines cross, and the slopes and y-intercepts tell you whether they cross once, never, or everywhere.

1. What Is a System of Linear Equations?

The equation $x + 2y = 8$ is satisfied by many pairs of values — for instance $x = 0,\ y = 4$, or $x = 8,\ y = 0$, or $x = 4,\ y = 2$. A second equation, $2x - y = 1$, has many solutions of its own: $x = 0,\ y = -1$, or $x = 1,\ y = 1$, or $x = 3,\ y = 5$. Requiring a pair to satisfy *both* equations at the same time leaves only one pair: $x = 2$ and $y = 3$, since $2 + 2(3) = 2 + 6 = 8$ ✓ and $2(2) - 3 = 4 - 3 = 1$ ✓.

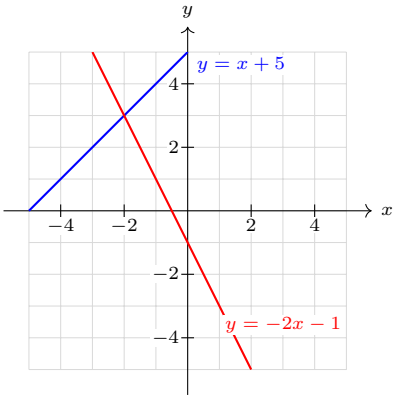
A **system of linear equations** is a set of two (or more) linear equations considered together. A **solution to the system** is an ordered pair (x, y) that satisfies every equation in the system simultaneously. Graphically, the solution is the **point of intersection** of the lines.

2. Reading a Solution From a Graph

When a system has already been graphed, its solution is read directly off the picture: it is the coordinates of the point where the two lines cross.

Example 1 — Reading the solution off a finished graph

The system $y = x + 5$ (blue) and $y = -2x - 1$ (red) has been graphed below. The system has an integer solution.



(a) State the solution: $x =$_____, $y =$_____

(b) Write the solution as an ordered pair.

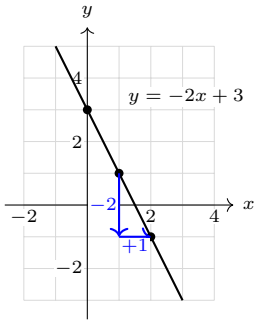
(c) Verify the solution by substituting it into both equations.

3. Solving a System by Graphing Without Technology

To solve a system by hand, graph each line — from a table of values, from the slope and y-intercept, or from the two intercepts — then read the coordinates of the point where the lines cross. Substituting those coordinates back into both original equations checks the solution: if both equations balance, the solution is confirmed. A hand-drawn graph gives an exact answer only when the lines cross on grid lines; when they cross between grid lines, the graph gives only an estimate (Lesson 2 finds exact values algebraically).

Graphing one line from slope–intercept form (from Unit 7). To graph $2x + y = 3$:

1. Rearrange into slope–intercept form: $y = -2x + 3$, so $m = -2$ and $b = 3$.
2. Plot the y-intercept $(0, 3)$.
3. Use the slope $m = \dfrac{-2}{1}$: from $(0, 3)$ go down 2 and right 1 to $(1, 1)$, then again to $(2, -1)$. Draw the line through the points.

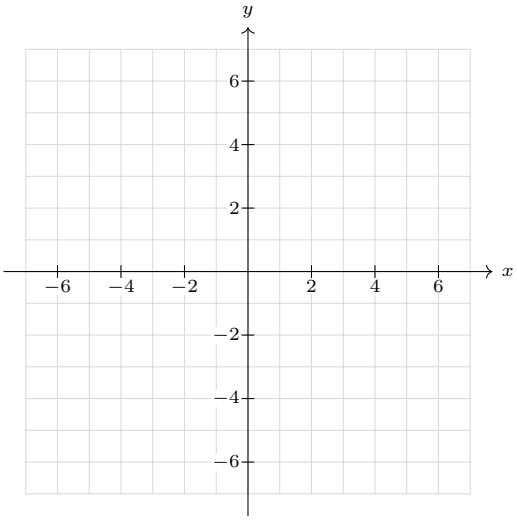


Shortcut when both equations are already in the form $y = \ldots$: at the crossing point both lines have the same y, so the two expressions can be set equal. For $y = x + 1$ and $y = -2x + 7$: $x + 1 = -2x + 7 \Rightarrow 3x = 6 \Rightarrow x = 2$, and then $y = 2 + 1 = 3$. The solution is $(2, 3)$.

Example 2 — Graph both lines, read the crossing point, verify it

Consider the system of equations $x - y = 2$ and $3x + y = -10$.

(a) Graph both equations on the grid below without using technology.



(b) State the solution to the system.

(c) Verify the solution algebraically by substituting the values into both original equations.

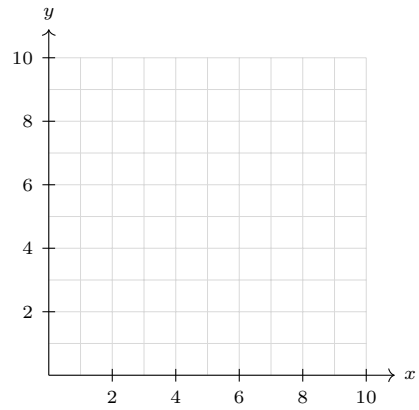
4. Using a System to Model a Real Situation

Many real situations involve two unknown quantities linked by two separate conditions. Writing one equation for each condition produces a system whose solution answers the original question.

Example 3 — Two families, two ticket prices

The Alvarez family pays \$9 in entry fees for 2 adult tickets and 1 child ticket to a charity fun run. The Bennett family pays \$7 in entry fees for 1 adult ticket and 3 child tickets. Let x be the entry fee (in dollars) for an adult ticket and y the entry fee (in dollars) for a child ticket, so that the Alvarez fees are modelled by $2x + y = 9$ and the Bennett fees by $x + 3y = 7$.

(a) Graph both equations without technology.



(b) State the coordinates of the point of intersection, and explain what the coordinates represent in the context of the question.

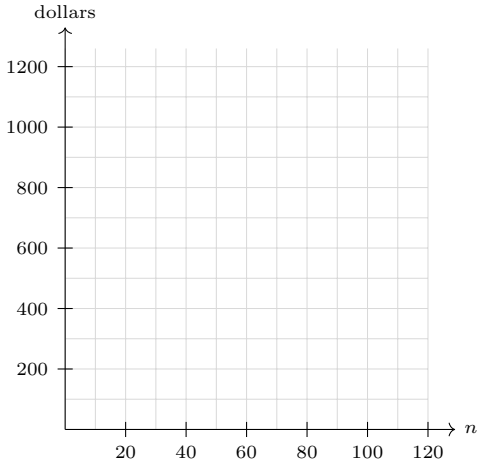
Break-even. A business **breaks even** when its cost and its revenue are equal — the point where the cost line and the revenue line cross. Where the revenue line is above the cost line the business makes a **profit**; where it is below, a **loss**. Both equations are written as "dollars = …", so the break-even point can also be found by setting the two expressions equal (the shortcut from Section 3).

Example 4 — Cost, revenue and the break-even point

The Robotics Club is fundraising by selling custom mugs. Designing and printing the mugs has a fixed setup cost of \$450, plus \$3 per mug produced. The club plans to sell the mugs for \$9 each. The cost C and the revenue R, both in dollars, from producing and selling n mugs are

$$\text{Cost: } C = 450 + 3n \qquad \text{Revenue: } R = 9n.$$

(a) Graph each equation on the grid provided.



(b) Mark the break-even point on the grid. Then confirm it algebraically by setting the cost equal to the revenue.

(c) How much profit or loss is made if **(i)** 30 mugs are sold? **(ii)** 90 mugs are sold?

(d) Shade the region of the graph where the club makes a profit.

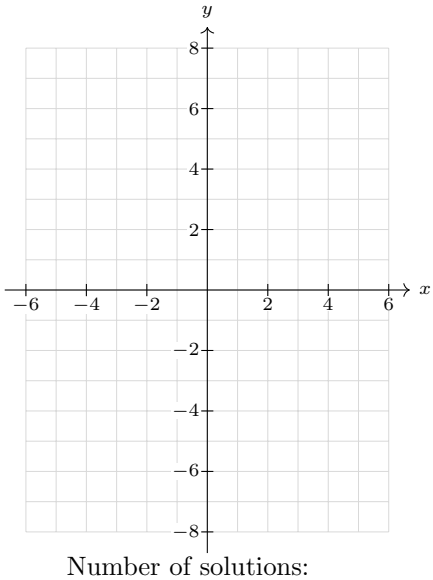
5. Three Ways Two Lines Can Meet

So far every system has had exactly one solution: the single point where the two lines cross. Two lines do not always cross exactly once, though. When two linear equations are graphed on the same set of axes, exactly one of three things happens: the lines cross at a single point, the lines run side by side and never touch, or the lines lie perfectly on top of one another. Each outcome gives the system a different number of solutions.

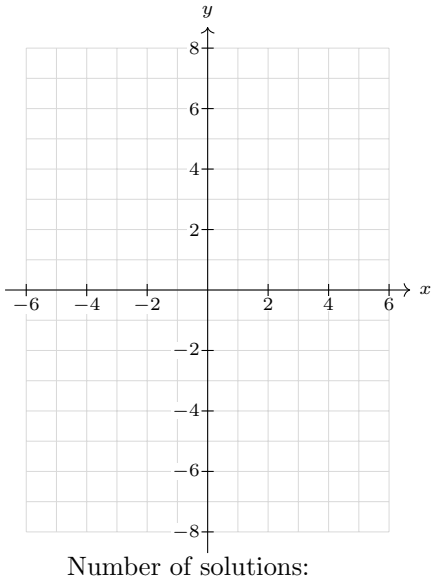
Example 5 — Graphing three systems and counting their solutions

Graph each system of equations on the grid provided, then state how many solutions the system has.

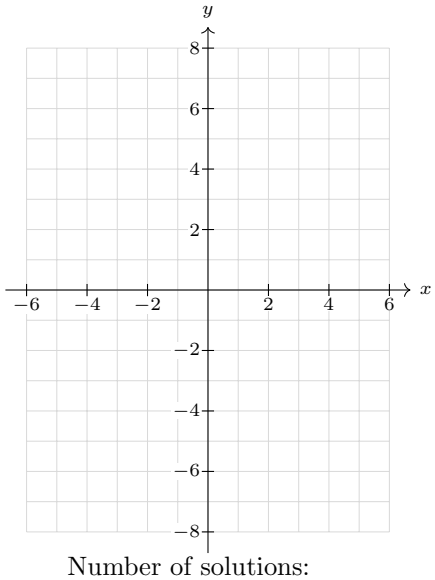
(a) $y = x - 2,\ \ y = -x + 6$ **(b)** $y = 2x + 3,\ \ y = 2x - 5$ **(c)** $y = 4x - 6,\ \ 8x - 2y - 12 = 0$



Number of solutions: _____



Number of solutions: _____



Number of solutions: _____

Look back at the slope and y-intercept of each pair of equations above, and compare them to what you actually drew on the three grids.

Example 6 — The classification chart

Use what you noticed in Example 5 to complete the chart below.

One Solution	No Solution	Infinitely Many
		
Slopes: _____ Intercepts: _____	Slopes: _____ Intercepts: _____	Slopes: _____ Intercepts: _____

6. Classifying a System Without Graphing

Once both equations of a system are written in slope–intercept form, the pattern from Example 6 predicts the number of solutions directly from the numbers in the equations — no graph paper required.

Example 7 — Classifying from $y = mx + b$ alone

Without graphing, determine whether each system has no solution, one solution, or infinitely many solutions.

(a) $5x + 2y = 20,\ \ y = -\frac{5}{2}x$ **(b)** $x - 3y + 9 = 0,\ \ y = \frac{1}{3}x + 3$ **(c)** $6x + y = 11,\ \ x - 5y = 20$

Lesson 2: Solving Systems of Equations by Substitution

An exact algebraic method: isolate one variable, substitute it into the other equation, solve, then back-substitute.

1. The Idea Behind Substitution

Reading a solution off a grid is only as precise as the grid, and it breaks down when the solution involves fractions. If one equation can be rearranged so that a single variable stands alone on one side, that expression can be dropped directly into the *other* equation in place of that variable, collapsing two equations in two unknowns into one equation in a single unknown.

Steps for Solving by Substitution

Step 1: Choose whichever equation is easier to rearrange, and solve it for one variable in terms of the other.
Step 2: Substitute that expression into the *other* equation. This produces one equation in a single variable.
Step 3: Solve the equation from Step 2 for the remaining variable.
Step 4: Substitute the value found in Step 3 back into the expression from Step 1 to find the value of the other variable.

Example 1 — One system, solved two different ways

Consider the following system of equations:

$$x + 3y = 13 \qquad 2x - y = 5$$

(a) Solve the system using the method of substitution by rewriting the first equation in the form $x = \ldots$.

(b) Solve the *same* system again, this time rewriting the first equation in the form $y = \ldots$. You should arrive at the same solution.

(c) Verify that your solution satisfies both original equations.

Example 2 — A rearrangement that produces a fraction

Consider the following system of equations:

$$2x + 3y = 0 \qquad 4x - 5y = 22$$

Solve the system using the method of substitution, and verify that your solution satisfies both equations. (Notice that the first equation has no constant term — rearranging it will produce a fraction, and that is perfectly normal.)

Example 3 — Exact values where a graph can only estimate

Graphed by hand, the lines $4x + 3y = 2$ and $2x - y = 5$ cross between grid lines, so the graph can only give an estimate of the solution (Lesson 1). Substitution gives the solution as exact values.

(a) Rewrite $2x - y = 5$ in the form $y = \ldots$

(b) Solve the system by substitution, and express the solution using exact values (fractions, not decimals).

(c) Verify the solution by substituting into both original equations.

2. Substitution When an Equation Must Be Expanded First

Sometimes brackets have to be expanded and like terms collected *before* either equation can be solved for one variable in terms of the other. Always clear the brackets first, then proceed with the same four steps.

Example 4 — Expanding before substituting

Consider the following system of equations:

$$a - 2(b - 3) = 8 \qquad 4a + 3(b - 2) = 13$$

(a) Expand both equations, then solve the system using the method of substitution.

(b) Verify algebraically that your solution satisfies both of the *original* (unexpanded) equations.

When both variables disappear: $0 = 0$ **or** $0 = k$

Sometimes substituting makes *both* variables cancel. What is left tells you the number of solutions — the same three cases as the Lesson 1 chart.

(i) $y = 3x - 1$ and $6x - 2y = 2$

$6x - 2(3x - 1) = 2$

$6x - 6x + 2 = 2$

$2 = 2$, i.e. $0 = 0$ — **always true**.

Every x works: **infinitely many solutions**. $6x - 2y = 2$ is $y = 3x - 1$ again — the **same line** (equal slopes, equal intercepts).

(ii) $y = 3x - 1$ and $6x - 2y = 8$

$6x - 2(3x - 1) = 8$

$6x - 6x + 2 = 8$

$2 = 8$, i.e. $0 = 6$ — **never true**.

No x works: **no solution**. $6x - 2y = 8$ is $y = 3x - 4$ — **parallel lines** (equal slopes 3, different intercepts -1 and -4).

3. Extension — Three Equations in Three Unknowns

Example 5 — (Extension) Three equations in three unknowns

Solve the following system using substitution.

$$4p - 3q - 2r = -9 \qquad r = 2p \qquad 3q - p - 5 = 0$$

Hint: the second and third equations already give r and q each in terms of p alone. Substitute both into the first equation to obtain a single equation in p.

Lesson 3: Combining Equations: The Elimination Method

Adding or subtracting two equations to remove a variable — directly, after scaling one equation, or after scaling both.

1. The Idea Behind Elimination

Adding equal amounts to equal amounts keeps them equal. Each equation states that its left side and its right side are the same number, so the two left sides may be added and the two right sides added, and the result is still true. If one variable has **opposite** coefficients, adding makes it vanish; if it has **matching** coefficients, subtracting does. Either way one equation in one variable is left.

Steps of the Elimination Method

Step 1: If necessary, multiply one or both equations by a constant so that one variable has matching or opposite coefficients in the two equations.

Step 2: Add the equations (opposite coefficients) or subtract them (matching coefficients) to eliminate that variable.

Step 3: Solve the resulting one-variable equation.

Step 4: Substitute that value back into either original equation to find the other variable, then verify the solution in *both* original equations.

Two cautions run through the whole lesson. Multiplying an equation by a constant means multiplying *every* term, the constant term included. Subtracting an equation reverses the sign of *every* term being subtracted, again including the constant — writing it as a bracketed difference, $(\cdots) - (\cdots)$, keeps the signs straight.

2. Eliminating by Adding or Subtracting Directly

When a system already has matching or opposite coefficients, Step 1 needs no work at all.

Example 1 — Opposite coefficients: add the equations

Consider the system $4x + 3y = 18$ $2x - 3y = 0$.

(a) Add the two equations together. Which variable is eliminated, and why?

(b) Solve the equation from part (a) for x, then substitute back into either original equation to find y.

(c) Verify that your solution satisfies both original equations.

Example 2 — Matching coefficients: subtract the equations

Consider the system $3x + 4y = 7$ $3x - 2y = 19$.

(a) Subtract the second equation from the first (Eq. 1 − Eq. 2). Which variable is eliminated, and why?

(b) Solve the equation from part (a) for y, then substitute back into either original equation to find x.

(c) Verify that your solution satisfies both original equations.

Choosing which variable to eliminate. When either would do, pick the one with the simpler coefficients — already equal or opposite, or small numbers — since that keeps the arithmetic in the later steps simple.

3. When One Equation Needs to Be Scaled

If neither variable has matching or opposite coefficients as the system stands, multiplying just one of the two equations by a constant may create a matching pair.

Example 3 — Scaling one equation

Consider the system $3x + 2y = 20$ $5x - y = 29$.

(a) As written, does adding or subtracting the equations eliminate either variable? Explain.

(b) Multiply Eq. 2 by a constant so that adding it to Eq. 1 eliminates y. Add, and solve for x.

(c) Substitute your x into either original equation to find y, then verify in both.

4. When Both Equations Need to Be Scaled

If no single equation can be scaled to match the other, multiply *both* equations, by possibly different constants, so that one variable ends up with matching or opposite coefficients. The smallest such pair comes from the least common multiple of the two coefficients.

Example 4 — When scaling one equation is not enough

Return to the system of Example 3: $3x + 2y = 20$ $5x - y = 29$.

(a) Can x be eliminated by multiplying only *one* equation by a whole number? Explain.

(b) Scale *both* equations so that x is eliminated, and solve for y.

Example 5 — Scaling both equations

Consider the system $4x + 5y = 1$ $3x - 2y = 18$.

(a) Choose whole numbers to multiply each equation by so that adding the results eliminates y.

(b) Add the equations from (a) and solve for x; substitute back for y and verify.

(c) Back to the original system: choose whole numbers so that x is eliminated instead.

(d) Combine the equations from (c) and solve for y.

5. Preparing a System Before Eliminating

Not every system arrives ready to eliminate: an equation may first need rearranging into the form $Ax + By = C$ (general form, $Ax + By + C = 0$, with the constant term moved to the other side), and fractions are cleared by multiplying through by a common denominator.

Example 6 — Rearranging into the form $Ax + By = C$; clearing fractions

Neither system below is ready to eliminate as it stands.

(a) Rearrange $5x - 2y - 7 = 0$, $2x = 3y - 6$ into the form $Ax + By = C$, solve by elimination, and verify in both originals.

(b) Clear the fractions in $\dfrac{3x - 5y}{4} = \dfrac{1}{2}$, $\dfrac{x}{3} + \dfrac{y}{2} = \dfrac{7}{3}$ and solve by elimination.

Special cases. If eliminating one variable removes *both* (as in Lesson 2's box), read the result the same way: a true statement such as $0 = 0$ means infinitely many solutions (the same line); a false statement such as $0 = 10$ means no solution (parallel lines). For example, doubling $x + 2y = 4$ and subtracting $2x + 4y = -2$ gives $(2x + 4y) - (2x + 4y) = 8 - (-2)$, that is, $0 = 10$.

6. Choosing a Method

Method	Use it when ...	Example
Graphing (by hand)	you need a picture or an estimate. A hand-drawn graph is exact only when the solution has integer coordinates.	$y = 2x - 1$, $y = -x + 5$
Substitution	a variable already has coefficient 1 or -1, or is already isolated ($y = \ldots$ or $x = \ldots$).	$y = 3x - 2$, $5x - 2y = 1$
Elimination	the coefficients of one variable already match or are opposites, or are easy multiples of each other.	$4x + 3y = 10$, $4x - 5y = -6$

Whichever method you choose, **verify** the solution in *both* original equations.

Lesson 4: Solving Word Problems with Systems of Equations

Translating a situation described in words into two equations, solving, and interpreting the answer in context.

1. A Four-Step Strategy for Word Problems

Almost every application problem in this lesson can be attacked with the same four steps, whether the situation involves numbers, geometry, money, or rates.

Steps for Solving an Application Problem

Step 1: Introduce variables to represent the unknown quantities. Say what each one stands for *and in what units.*
Step 2: Use the given information to form a system of two equations in those variables.
Step 3: Solve the system, using graphing, substitution, or elimination.
Step 4: Answer the original question in words, and check that your solution makes sense in the context of the problem.

Step 3: choosing a method (the table from Lesson 3, Section 6):

Graphing	a picture or estimate is enough (by hand: exact only for integer solutions)
Substitution	a variable already has coefficient 1 or -1, or is isolated — e.g. $f + t = 23 \Rightarrow f = 23 - t$
Elimination	the coefficients of one variable already match or are opposites, or are easy multiples

Step 4 is not optional. A system can be solved correctly and still answer the wrong question, so check the solution against the *words* as well as against the equations: totals must add up, counts of objects must be whole numbers, and prices, lengths and times must be positive.

2. Number Relationships

Some problems describe a relationship between two unknown numbers with no physical context at all. The words *sum*, *difference*, *more than* and *twice* translate directly into algebraic expressions.

Example 1 — Two numbers, a difference and a "more than" relationship

The difference between two numbers is 11. The larger number is 2 more than twice the smaller number.

(a) Introduce variables for the larger number and the smaller number.

(b) Use the given information to form a system of two equations.

(c) Solve the system to find both numbers.

(d) State the answer in words, and verify that both numbers satisfy the original conditions.

3. Perimeter and Dimension Problems

When a problem gives the perimeter of a rectangle together with a relationship between its length and width, the perimeter formula and the length–width relationship become the two equations.

Example 2 — Perimeter and a length–width relationship

A rectangular garden has a perimeter of 54 metres. Its length is 3 metres more than twice its width. Find the dimensions of the garden.

4. Money and Denomination Problems

Here one equation counts *how many* items there are and the other totals their *value*. Keeping the two straight — one for a count, one for a value — is the key to setting up the system correctly.

Example 3 — One equation counts, the other totals value

Maria has a total of \$310 in five-dollar bills and twenty-dollar bills. If she has 23 bills in total, how many of each denomination does she have?

5. Investment and Interest Problems

Investment problems work the same way, except the two equations total the *principal* invested and the *interest* earned. Interest rates given as percentages must be converted to decimals before they are used in an equation.

Example 4 — One equation totals principal, the other totals interest

Devon invested his savings of \$36 000 in two accounts. One account earned 8% interest and the other earned 5% interest. If he earned a total of \$2340 in interest for the year, how much did he invest at each rate?

6. Rate and Production Problems

A final common type gives two different rates — two machines producing items at different speeds, say — together with a combined total. One equation totals the *time*, the other totals the *output*.

Example 5 — One equation totals time, the other totals output

A factory has an older machine that packs 25 boxes per hour and a newer machine that packs 35 boxes per hour. Yesterday the two machines ran for a combined total of 18 hours and packed 550 boxes. For how many hours did each machine run?

Lesson 5: Mixture and Motion: More Applications of Linear Systems

Blending two ingredients that differ in cost or concentration, and journeys made of two legs or helped and hindered by a current or wind.

1. Mixture Applications

A mixture problem combines two ingredients — candies, chemical solutions, metals — that differ in some rate, such as price per kilogram or percent concentration. Two equations are always available: one adds up the **total amount** of the mixture (its mass, volume, or number of items), the other adds up the **total value** that amount carries (the total cost, or the total salt, gold, or alcohol content). Write both equations down before solving; the system then yields to substitution or elimination exactly as in earlier lessons.

Two habits keep mixture work honest. Define each variable with its **unit** ("x = mL of the 30% solution"), so that litres of *solution* are never confused with litres of *pure acid*. And clear percent rates to integers before solving — multiplying every term of $0.30x + 0.80y = 125$ by 10 gives $3x + 8y = 1250$, which is easier to eliminate with.

Mixture organizer	**Amount**	**× Rate**	**= Value**
Ingredient A	x	rate A	(rate A) x
Ingredient B	y	rate B	(rate B) y
Mixture	total amount	mixture rate	total value

The **Amount** column gives one equation (the two amounts add to the total amount); the **Value** column gives the other (the two values add to the total value). Rates are never added.

Example 1 — Blending by cost per kilogram

A café blends premium coffee beans costing \$24/kg with standard coffee beans costing \$14/kg. The blend weighs 60 kg in total and sells for \$17/kg.

	Amount (kg)	**× Rate (\$/kg)**	**= Value (\$)**
Premium beans			
Standard beans			
Blend			

(a) Let p be the kg of premium beans and s the kg of standard beans. Complete the organizer, then write a system of two equations — one from the Amount column (total mass), one from the Value column (total value).

(b) Solve the system from part (a) to find p and s.

(c) State how many kilograms of each bean are in the blend, and verify the solution in both original equations.

Example 2 — Blending by percent concentration

A lab technician needs 250 mL of a 50% saline solution. She has a stock solution that is 30% saline and another that is 80% saline.

(a) Let x be the mL of 30% solution and y the mL of 80% solution. Write a system of two equations — one for the total volume, one for the total amount of salt.

(b) Solve the system from part (a) to find x and y.

(c) State how many mL of each stock solution are required, and verify the solution.

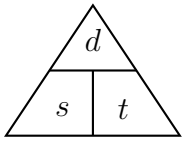
2. Distance, Speed, and Time Applications

Distance, speed, and time are related by $d = s\,t$, which rearranges to $s = \frac{d}{t}$ and $t = \frac{d}{s}$. The triangle is a memory aid: cover the quantity you want, and what is left shows how to build it from the other two.

Most applications split a journey into two **legs** — two different speeds, or a trip out and a trip back — and the two equations come from a relationship between the two distances, the two times, or the two speeds. A current or a wind that helps on one leg and hinders on the other is the standard case: if b is the speed in still water or still air and c is the speed of the current or wind, the ground speed is $b + c$ going with it and $b - c$ going against it. Stopped time is not driving time; subtract any breaks from the total before writing the time equation.

Motion organizer	**Distance**	**= Speed**	**× Time**
Leg 1			
Leg 2			
Total	total distance	—	total time

Each leg's row must satisfy $d = s \times t$. In the Total row, distances add and times add — speeds do not.



Example 3 — Two speeds on one trip

A trucker drove 810 km from Edmonton to Regina in 11.5 hours total, including a 1-hour fuel stop and a 30-minute rest break. While actually driving she averaged 95 km/h on the highway and 60 km/h on secondary roads.

	Distance (km)	**= Speed (km/h)**	**× Time (h)**
Highway			
Secondary roads			
Total		—	

(a) Let h be the hours driven on the highway and r the hours driven on secondary roads. Complete the organizer, then write a system of two equations. (Hint: find the total *driving* time first by subtracting the stops.)

(b) Solve the system from part (a) to find h and r.

(c) State how much time she spent driving on the highway, and verify the solution.

Example 4 — Downstream and upstream

A ferry takes 4 hours to travel 96 km downstream (with the current) and 6 hours to travel the same 96 km upstream (against the current).

(a) Let b be the ferry's speed in still water (km/h) and c the speed of the current (km/h). Using speed = distance ÷ time, write a system of two equations.

(b) Solve the system from part (a) to find b and c.

(c) State the speed of the current and the ferry's speed in still water, and verify the solution.

Example 5 — Headwind and tailwind

A small plane flying against a headwind takes 2.5 hours to fly the 525 km from Calgary to Saskatoon. On the return trip, with the same wind now behind it, the plane takes only 1.75 hours to cover the same distance.

(a) Let p be the plane's speed in still air (km/h) and w the speed of the wind (km/h). Write a system of two equations.

(b) Solve the system from part (a) to find p and w.

(c) State the speed of the wind and the plane's speed in still air, and verify the solution.

Lesson 6: Systems of Linear Equations Unit Review

Every method of the unit on one page — classifying, graphing, substitution, elimination, and choosing among them — then mixed examples.

1. Method Summary

How many solutions? (Lesson 1, plus what the algebra shows in Lessons 2–3)

	One solution	**No solution**	**Infinitely many**
Graph	 lines cross	 parallel	 same line
Slopes / y-intercepts	different slopes	equal slopes, different intercepts	equal slopes, equal intercepts
Substitution or elimination gives	one value for each variable	a false statement, e.g. $0 = 5$	a true statement, e.g. $0 = 0$

Graphing (Lesson 1)

By hand: rearrange to $y = mx + b$, plot b, use the slope; read the crossing point.

Shortcut: both equations in the form $y = \ldots$? Set the two expressions equal.

Verify in both equations.

Substitution (Lesson 2)

1. Solve one equation for one variable.
2. Substitute into the *other* equation.
3. Solve for the remaining variable.
4. Back-substitute; verify in both.

Elimination (Lesson 3)

1. Scale one or both equations (every term) so one variable has matching or opposite coefficients.
2. Subtract (matching) or add (opposite).
3. Solve; back-substitute; verify in both.

Choosing a method (Lesson 3, Section 6)

Graphing	you need a picture or an estimate (by hand: exact only for integer solutions)
Substitution	a variable already has coefficient 1 or -1, or is already isolated
Elimination	the coefficients of one variable already match or are opposites, or are easy multiples

2. Mixed Examples

Example 1 — From two tables to the point of intersection

Two hikers each recorded a table of values while tracking their own straight-line trail marker on a coordinate grid.

Priya's line

x	y
-3	-7
-1	-3
1	1
3	5
5	9

Devon's line

x	y
-2	10
0	8
2	6
4	4
6	2

(a) Use each table to determine that hiker's equation, in the form $y = mx + b$.

(b) The two tables share no ordered pair. Determine the point where the two lines intersect, and state the value of $x + y$ at that point.

Example 2 — Choose a method, justify it, solve

For each system, name the method you would use and give a reason, then solve the system and verify.

(a) $y = 3x - 2,\ \ 5x - 2y = 1$ **(b)** $4x + 3y = 10,\ \ 4x - 5y = -6$ **(c)** $y = 0.8x + 1.2,\ \ y = -1.5x + 6.95$

Example 3 — A trip out and a trip back

(Multiple Choice) Sam walks at 6 km/h and runs at 10 km/h. Going from his house to school he walks part of the way and runs the rest, and the trip takes 42 minutes. Coming home he runs *twice* as far as he ran going to school (and walks the remainder); this trip takes 34 minutes. The distance between Sam's house and school is

A. 4 km B. 5 km C. 6 km D. 7 km

Example 4 — Comparing three plans

A phone carrier offers three data plans. Plan 1 costs \$25 per month plus \$0.50 per GB. Plan 2 costs \$18 per month plus \$0.90 per GB. Plan 3 costs \$70 per month for unlimited data. Let y = total cost per month in dollars and x = number of GB of data used per month.

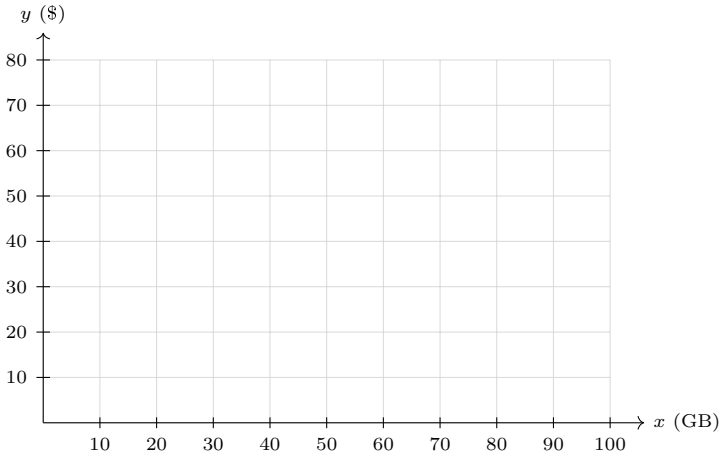
(a) What factor determines which plan is most economical for a given customer?

(b) Write a linear equation for each of the three plans.

(c) Determine algebraically the number of GB at which Plans 1 and 2 cost the same amount, and state that common cost.

(d) For each of Plans 1 and 2, determine the number of GB that could be purchased for \$70.

(f) Graph all three plans on the grid and mark the two crossover points where the cheapest plan changes.



(e) Devise a simple rule that determines which plan is most economical, depending on the expected number of GB of data used per month. Use the graph in (f).